CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First/Second Semester of B. Tech. Examination (EC/ME/CL/CE/IT/EE)

May 2012

PY 101 Engineering Physics

Date: 17.05.2012, Thursday | **Time: 10:00 a.m. To 01:00 p.m.** | **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 Choose the correct answer for the following questions. **[07]**

(a) If A = 1.0 ± 0.2 *m* and B = 2.0 ± 0.2 *m*, what is the square root of (A × B)?

(i) 1.4 ± 0.1 *m* (ii) 1.4 ± 0.2 *m*

(iii) 1.4 ± 0.4 *m* (iv) 1.4 ± 0.5 *m*

(b) L = 0.020540 *m* have__________ significant digits.

(i) 6 (ii) 5

(iii) 4 (iv) 3

(c) What is the value of intensity level of threshold of feeling?

(i) 120 *dB* (ii) 60 *dB*

(iii) 120 *bel* (iv) 12 *dB*

(d) A train whistle has an acoustic power output of 100 *W*. If the sound energy spreads out spherically, what is the intensity level in dB at a distance of 100 *m* from the train? ($I_0 = 10^{-12}\ W\,m^{-2}$).

(i) 55 *dB* (ii) 89 *dB*

(iii) 95 *dB* (iv) 120 *dB*

(e) When certain light rays pass from a vacuum into a block of an unknown material. The measured index of refraction of the material is 3.50. What is the speed of light inside the block?

(i) $1.0 \times 10^7\ m/s$ (ii) $4.8 \times 10^7\ m/s$

(iii) $8.6 \times 10^7\ m/s$ (iv) $1.9 \times 10^8\ m/s$

(f) The persistence of audible sound due to the successive reflections from the surrounding objects even after the source has stopped to produce sound is called ________.

(i) Reflection (ii) Reverberation

(iii) Echo (iv) Echelon

(g) He-Ne laser gives_______ colored light.

(i) Red (ii) Green

(iii) Orange (iv) Blue

Q - 2 (a) What do you mean by error in a measurement? Explain why should we care about errors? [3]

(b) A hall has a volume of 125000 m^3 and reverberation time of 2.2 *sec*. If 2000 cushioned chairs are additionally placed in the hall, what will be the new reverberation time of the hall? The absorption of each chair is 1.1 $Sabine.m^2$. [4]

(c) Discuss the various factors affecting the acoustics of building and their remedies. [7]

OR

Q - 2 (a) Define Pitch, Loudness and Timbre. [3]

(b) An ultrasonic beam of 1.0 *cm* wavelength sent by ship returns from an iceberg after 2 *seconds*. If velocity of ultrasonic beam in sea water is 1510 *m/s* at 0^0 *C*. Find the frequency of ultrasound and the distance between iceberg and the ship. [4]

(c) What is electrostriction effect? How this effect is utilized for the production of ultrasonic waves? [7]

Q - 3 (a) Distinguish between laser light and ordinary light. [3]

(b) Give the expression for the ratio of spontaneous and stimulated emission. The ratio of population of two energy levels at 300 *K* is 10^{-30}. Find the wavelength of the radiation emitted. [4]

(c) Derive the expression for acceptance angle and numerical aperture of an optical fiber. In a step index fiber, relative index difference is 2 % and cladding refractive index is 1.40. Find (i) core refractive index and (ii) acceptance angle. [7]

OR

Q - 3 (a) Define: (i) total internal reflection, (ii) metastable state and (iii) modal dispersion. [3]

(b) List out four differences between single mode fiber and multimode fiber. [4]

(c) What is the active medium in Nd:YAG LASER? Explain with the energy level diagram, the construction & working of Nd:YAG LASER. [7]

SECTION – II

Q - 4 Choose the correct answer for the following questions. **[07]**

(a) Superconductivity is a phenomenon where it is observed that at very low temperature some metals and compounds acquire zero electrical resistance and

(i) Zero dielectric strength (ii) Zero magnetic induction

(iii) Zero temperature coefficient (iv) All the above

(b) Maglev trains are constructed based on _______ effect.

(i) Gravitational (ii) Meissner

(iii) Electrical (iv) none

(c) The number of atoms per unit cell of BCC structure is _______.

(i) 1 (ii) 2

(iii) 3 (iv) 4

(d) What is the operating voltage of the X-ray tube to produce X-rays of wavelength 0.02 *nm?*

(i) 62 kV (ii) 620 kV

(iii) 6.20 kV (iv) 62000 kV

(e) Which of the following types of electromagnetic wave has the highest wavelength?

(i) x-rays (ii) visible light

(ii) ultraviolet (iv) gamma rays

(f) The number of Bravais lattice in cubic crystal system is ________.

(i) 1 (ii) 2

(iii) 3 (iv) 4

(g) Nanotechnology rests on the technology that involves devices and systems:

(i) 1- 100 nm (ii) 1-200 nm

(iii) up to 500 nm (iv) up to 1000 nm

Q – 5 (a) Define Hall coefficient. What is the mobility of carriers if its resistivity is $8.93\times10^{-3}\ \Omega m$ and R_H of the specimen is $3.66\times10^{-4}\ m^3/C$. [3]

(b) Compare type-I and type-II superconductors (four points). [4]

(c) Write any three applications of X-rays. A voltage of 25 kV is applied across the X-ray tube and a current of 10 mA flows through it. Find the number of electrons striking the target per second and the minimum wavelength in the X-ray spectrum. [7]

OR

Q - 5 (a) What is the effect of magnetic field on a super conductor? Hence explain Meissner effect. [3]

(b) What is SQUID? Explain the formation of SQUID and its applications. [4]

(c) Distinguish between conductors and superconductors. List any five applications of super conductors. [7]

Q - 6 (a) Define Lattice plane, unit cell and primitive cell. [3]

(b) What is interplanar distance? Calculate the atomic radius if the interplanar spacing of (110) plane is $2\,A^0$ for a cubic crystal. [4]

(c) What is nanotechnology? Write any four important scopes of applications of nanotechnology in the field of engineering and technology. [7]

OR

Q - 6 (a) Sketch the following Miller indices planes of a cubic cell: (002), (110) and (111). [3]

(b) What is X-ray crystallography? A beam of X-rays is incident on a NaCl crystal with lattice plane spacing 0.282 *nm*. Calculate the wavelength of X –rays if the first order Bragg reflection taken place at a grating angle 8.58^0. [4]

(c) Distinguish between bulk and nanomaterials. Write any five size dependent properties of objects in nanoscale. [7]

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